1.

(1) $G_Z(z) = G_X(z)G_Y(z) = \mathrm{e}^{\lambda_1(z-1)}\mathrm{e}^{\lambda_2(z-1)} = \mathrm{e}^{(\lambda_1+\lambda_2)(z-1)}$, 即 $Z \sim \mathrm{Poisson}(\lambda_1+\lambda_2)$.

(2)

$$\begin{aligned}
\Pr[X=k \mid Z=n] &= \frac{\Pr[X=k \wedge Z=n]}{\Pr[Z=n]} \\
&= \frac{\Pr[Y=n-k]\Pr[X=k]}{\Pr[Z=n]} \\
&= \frac{\frac{\mathrm{e}^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}\frac{\mathrm{e}^{-\lambda_1}\lambda_1^{k}}{k!}}{\frac{\mathrm{e}^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}} \\
&= \frac{\binom{n}{k}\lambda_1^k\lambda_2^{n-k}}{(\lambda_1+\lambda_2)^n}.
\end{aligned}$$

2.

记 $n = 2024$. 容斥给出:

$$\begin{aligned}
\Pr\left[\bigvee_{i=1}^{n} E_i\right] &= \sum_{\varnothing \neq S \subset [1:n]} (-1)^{|S|-1}\Pr\left[\bigwedge_{i\in S} E_i\right] \\
&= \sum_{i=1}^{n}\binom{n}{i}(-1)^{i+1}\frac{1}{i+1} \\
&= \left(\int((1+x)^n-1)\mathrm{d}x\right)\Bigg|_0^{-1} \\
&= \frac{n}{n+1} = \frac{2024}{2025}.
\end{aligned}$$

3.

令 $p = \frac{1}{2}$ 表示单次掷出 H 的概率.

(1)

$$\begin{aligned}
G_{T_{\mathrm{HHH}}}(z) &= p^3z^3 + p^2(1-p)z^3G_{T_{\mathrm{HHH}}}(z) + p(1-p)z^2G_{T_{\mathrm{HHH}}}(z) + (1-p)zG_{T_{\mathrm{HHH}}}(z) \\
&= \frac{p^3z^3}{1-p^2(1-p)z^3-p(1-p)z^2-(1-p)z}.
\end{aligned}$$

则

$$\mathbb{E}[T_{\mathrm{HHH}}] = G'_{T_{\mathrm{HHH}}}(1) = 14.$$

(2)

$$\begin{aligned}
G_{T_{\mathrm{HHT}}}(z) &= \sum_{i=2}^{+\infty} p^i(1-p)z^{i+1} + p(1-p)z^2G_{T_{\mathrm{HHT}}}(z) + (1-p)zG_{T_{\mathrm{HHT}}}(z) \\
&= \frac{p^2(1-p)z^3}{(1-p(1-p)z^2-(1-p)z)(1-pz)}.
\end{aligned}$$

则

$$\mathbb{E}[T_{\mathrm{HHT}}] = G'_{T_{\mathrm{HHT}}}(1) = 8.$$

(3) 考虑在匹配 HTH 的 DFA 上的 Markov 过程. 设 $\mathbb{E}[\mathrm{str}]$ 表示从字符串 str 对应的 DFA 状态走到接受状态的期望步数,

$$\begin{cases} \mathbb{E}[\varnothing] = 1 + (1-p)\mathbb{E}[\varnothing] + p\mathbb{E}[\mathrm{H}], \\ \mathbb{E}[\mathrm{H}] = 1 + p\mathbb{E}[\mathrm{H}] + (1-p)\mathbb{E}[\mathrm{HT}], \\ \mathbb{E}[\mathrm{HT}] = 1 + (1-p)\mathbb{E}[\varnothing]. \end{cases}$$

解得

$$\mathbb{E}[\varnothing] = \mathbb{E}[T_{\mathrm{HTH}}] = 10.$$

4.

设随机变量 $B \sim \mathrm{Bern}(1-\lambda)$, R 满足 $P_{R|B=0} = P$, $P_{R|B=1} = Q$, 恰好有 $P_R(x) = \lambda P(x) + (1-\lambda)Q(x)$, 这时

$$H[\lambda P + (1-\lambda)Q] = H[R] \geq H[R \mid B] = \lambda H[P] + (1-\lambda)H[Q].$$

5.

$$\begin{aligned} D(P_{XY} \parallel P_X P_Y) &= \mathbb{E}_{(X,Y)\sim P_{XY}}\left[\log \frac{P_{XY}(X,Y)}{P_X(X)P_Y(Y)}\right] \\ &= \mathbb{E}_{(X,Y)\sim P_{XY}}\left[\log \frac{1}{P_X(X)}\right] + \mathbb{E}_{(X,Y)\sim P_{XY}}\left[\log \frac{1}{\log P_Y(Y)}\right] \\ &\quad - \mathbb{E}_{(X,Y)\sim P_{XY}}\left[\log \frac{1}{P_{XY}(X,Y)}\right] \\ &= H[X] + H[Y] - H[X,Y] = I(X;Y). \end{aligned}$$

6.

不妨取 $\log = \ln$.

$$\begin{aligned} f(p,q) &:= d(p \parallel q) - 2(p-q)^2 \\ &= (1-p)\ln\frac{1-p}{1-q} + p\ln\frac{p}{q} - 2(p-q)^2 \\ &= (1-p)\ln(1-p) + p\ln p - (1-p)\ln(1-q) - p\ln q - 2(p-q)^2. \end{aligned}$$

显然 f 在定义域上的性质充分好. 计算

$$\begin{aligned} f'_q &= \frac{1-p}{1-q} - \frac{p}{q} + 4(p-q) \\ &= \frac{(q-p) + 4(p-q)q(1-q)}{(1-q)q} \\ &= \frac{(p-q)(4q(1-q)-1)}{(1-q)q}. \end{aligned}$$

则 $q = p$ 和 $q = \frac{1}{2}$ 时 f 可能有极值. $f(p,p) = 0$, 而

$$g(p) := f\left(p, \frac{1}{2}\right) = -h(p) + \ln 2 - 2\left(p - \frac{1}{2}\right)^2 \geq g\left(\frac{1}{2}\right) = 0.$$

所以 $f(p,q) \geq 0$, 明所欲证.

7.

已知

$$D(P_{XY} \parallel Q_{XY}) = D(P_X \parallel Q_X) + D\left(P_{Y|X} \parallel P_{Y|X} \middle| P_X\right) = D(P_X \parallel Q_X),$$

则显然

$$D(P_X \parallel Q_X) = D(P_{XY} \parallel Q_{XY}) \geq D(P_Y \parallel Q_Y).$$

8.

不妨取 $\log = \ln$, 题面中 P, Q 记为 P_X, Q_X. 欲证 $D(P_X \parallel Q_X) \geq 2\Delta(P_X, Q_X)^2$. 设 $S = \arg\max_E (P(E) - Q(E))^2$, 构造 $P_{Y|X=x}(y) = [y = [x \in S]]$, 那么 $P_Y = P_X \circ P_{Y|X} = \mathrm{Bern}(P_X(S))$, 同理 $Q_Y = \mathrm{Bern}(Q_X(S))$, 结合 6. 7. 的结论:

$$D(P_X \parallel Q_X) \geq D(P_Y \parallel Q_Y) = d(P_X(S), Q_X(S)) \geq 2(P_X(S) - Q_X(S))^2 = 2\Delta(P_X, Q_X).$$

9.

(1) 考虑

$$\begin{aligned} I[X;Z \mid Y] &= H[X \mid Y] + H[Z \mid Y] - H[XZ \mid Y] \\ &= H[XY] - H[Y] + H[YZ] - H[Y] - H[XYZ] + H[Y] \\ &= H[XY] + H[YZ] - H[XYZ] - H[Y] \\ &\geq 0. \end{aligned}$$

类似地:

$$\begin{cases} I[X;Z \mid Y] = H[XY] + H[YZ] - H[XYZ] - H[Y] \geq 0, \\ I[Z;Y \mid X] = H[ZX] + H[XY] - H[XYZ] - H[X] \geq 0, \\ I[X;Y \mid Z] = H[ZX] + H[YZ] - H[XYZ] - H[Z] \geq 0. \end{cases}$$

三式相加:

$$2\sum_{\text{cyc}} H[XY] - 3H[XYZ] - \sum_{\text{cyc}} H[X] \geq 0.$$

而显然 $\sum_{\text{cyc}} H[X] \geq H[XYZ]$, 所以

$$\begin{aligned} 2\sum_{\text{cyc}} H[XY] &\geq 3H[XYZ] + \sum_{\text{cyc}} H[X] \geq 4H[XYZ] \\ &\Rightarrow 2H[XYZ] \leq H[XY] + H[YZ] + H[ZX]. \end{aligned}$$

且取等当且仅当

$$\begin{cases} I[X;Z \mid Y] = I[Z;Y \mid X] = I[X;Y \mid Z] = 0, \\ H[X] + H[Y] + H[Z] = H[XYZ]. \end{cases}$$

(2) 设空间点 $\{S_i(x_i, y_i, z_i)\}_{i=1}^n$, 考虑联合分布 $P_{XYZ}(X = x_i, Y = y_i, Z = z_i) = \frac{1}{n}$, 则

$$\{|XY|, |YZ|, |ZX|\} = \{n_1, n_2, n_3\},$$

因而

$$\sum_i \log n_i \geq \sum_{\text{cyc}} H[XY] \geq 2H[XYZ] = 2\log n,$$

也即是

$$n^2 \leq n_1 n_2 n_3.$$

取等时, XY, YZ, ZX 各自是均匀分布, 且满足条件互信息为 0, 容易看出, 这当且仅当 $\{S_n\}$ 是一个满的空间网格 (不难反证验证), 即存在 $A, B, C \subset \mathbb{R}$, 使得 $\{S_n\} = A \times B \times C$.